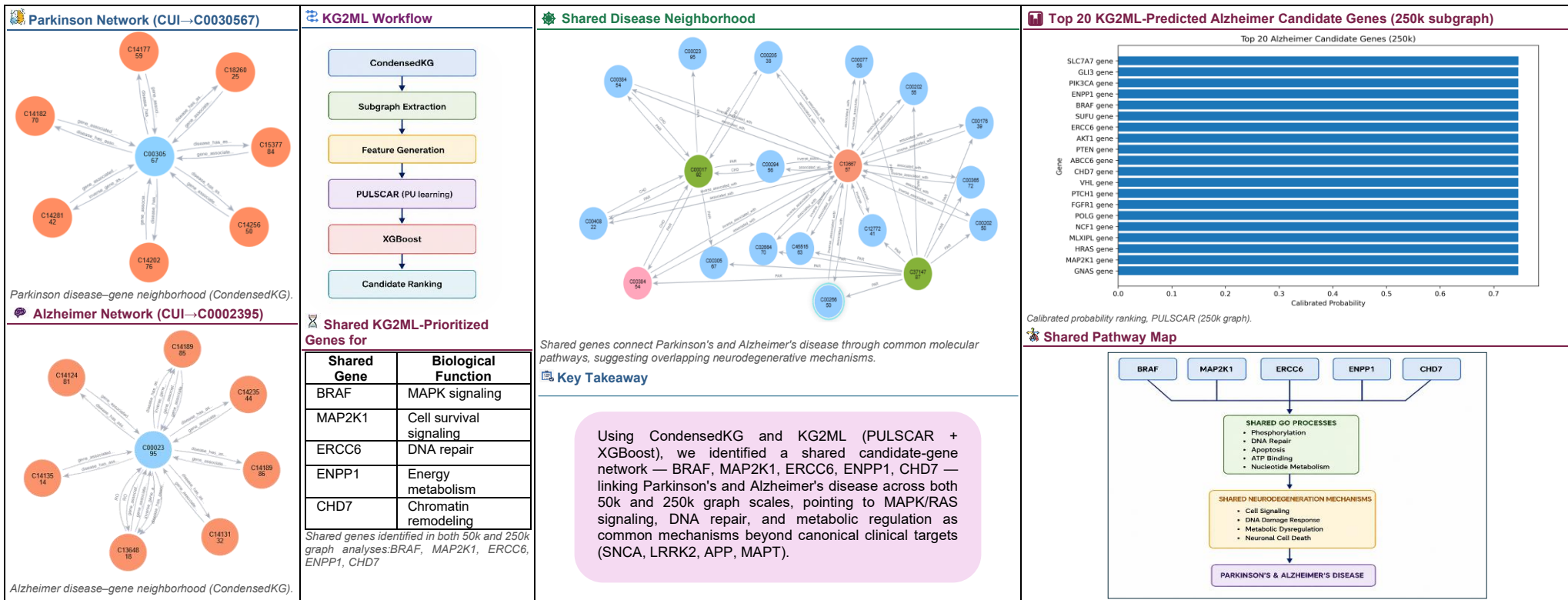


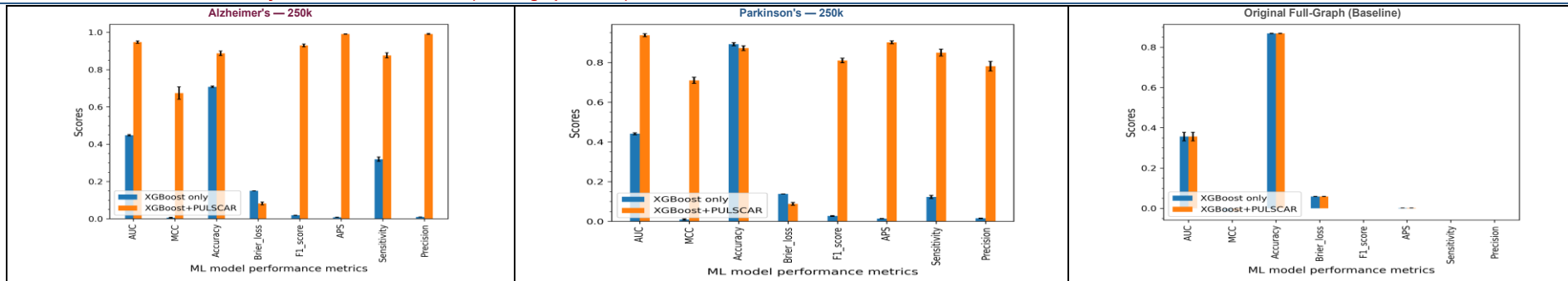
Decoding Neurodegeneration

Knowledge Graph–Based Discovery of Shared Molecular Mechanisms Linking Parkinson's and Alzheimer's Disease • CondensedKG + KG2ML + PULSCAR

Using CFDE CondensedKG and graph-based machine learning, we identified five shared candidate genes and common biological pathways linking Parkinson's and Alzheimer's disease.



Best Model Performance — XGBoost Only vs. XGBoost + PULSCAR (across graph scales)



Note: the baseline (original full-graph, pre-subgraph-optimization) run shows near-identical XGBoost-only vs. PULSCAR scores with weak separation — motivating the move to 250k disease-centered subgraphs, where PULSCAR clearly improves AUC, F1, APS, sensitivity, and precision.

Approx. Performance Metrics Summary (XGBoost + PULSCAR)						Disease & Graph-Scale Comparison from Parkinson's and Alzheimer's Top-20 candidate lists			
Run	AUC	MCC	F1	APS	Precision	Disease	Scale	Candidate Genes	Shared Candidates
Alzheimer 50k	~0.94	~0.68	~0.85	~0.96	~0.96	Parkinson's	50k	8,803	OCRL
Alzheimer 250k	~0.95	~0.67	~0.93	~0.99	~0.99	Parkinson's	250k	9,697	BRAF, MAP2K1, ERCC6, ENPP1, CHD7
Parkinson 50k	~0.90	~0.47	~0.89	~0.98	~0.99	Alzheimer's	50k	8,825	OCRL
Parkinson 250k	~0.92	~0.71	~0.81	~0.90	~0.78	Alzheimer's	250k	9,546	BRAF, MAP2K1, ERCC6, ENPP1, CHD7
Baseline (full graph)	~0.36	~0.00	~0.00	~0.00	~0.00				

What is a CUI?

A CUI (Concept Unique Identifier) is a permanent, vocabulary-independent UMLS code for one biomedical concept (a disease, gene, or compound), so the same concept can be matched across many source ontologies. Labels like C0030567 (Parkinson) and C0002395 (Alzheimer) in the network figures are CUIs.

Clinical Relevance

BRAF→MAPK signaling · MAP2K1→Cell survival · ERCC6→DNA repair · ENPP1→Energy metabolism · CHD7→Chromatin regulation